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📁 [Portfolio](#)

Summary

Final-year Computer Engineering undergraduate with production and research experience bridging **backend engineering**, **cloud/DevOps**, and **applied AI/ML**. Proven track record of shipping **Node.js** and **FastAPI** microservices and automated SaaS cron pipelines, alongside architecting advanced **RAG** systems, **multi-agent LLMs**, and zero-trust security frameworks. Highly proficient in secure deployments utilizing **Docker**, **Kubernetes**, **AWS EC2**, **CI/CD**, and hybrid **cryptography**. Seeking a full-time **Software Engineer** or **AI Engineer** role to build robust, scalable systems.

Technical Skills

Languages: Python, C++, JavaScript, TypeScript, Java

Frontend: React.js, Next.js, Angular, Tailwind CSS, Material-UI (MUI), Flutter, HTML/CSS

Backend & APIs: Node.js, Express.js, REST APIs, Firebase

AI / Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, RAG, Hugging Face Transformers, OpenAI, Prompt Engineering

Databases: MySQL, MongoDB, Redis, Cloud Firestore

Cloud & DevOps: Docker, Kubernetes, Jenkins, GitHub Actions, AWS EC2, Terraform, Ansible, CI/CD, Prometheus, Grafana

Systems / HPC: CUDA, OpenMP, MPI, Linux

Tools: Git, GitHub, Jira, Figma, Postman, Agile/Scrum

Experience

Global Wavenet (Pvt) Ltd

onsite

Software Engineer Intern

Jun 2025 – Dec 2025

- Developed and debugged backend services and API endpoints. Designed and implemented automated **Cron jobs** using **Node.js**, **JavaScript**, and **MySQL**, and tested REST APIs using Postman.
- Built and configured IVR and AI-powered conversational call flows using Wavenet Compose, integrating natural-language logic into voice automation systems.
- Gained applied experience with **Retrieval-Augmented Generation (RAG)**, **LangChain**, **LangGraph**, vector databases, and **Azure OpenAI** for conversational AI solutions.
- Containerized and deployed services with **Docker**, supporting build and release workflows across server environments.
- Contributed to frontend features using **Angular** and collaborated with cross-functional development
- deployed custom **Call Detail Record (CDR)** logging systems across multiple networks to precisely track AI call details

Projects

EDCF-Shield – Zero-Trust Security for SDVN | C++, LLM, TGNNs, Blockchain, NS-3 | [🐙Code](#)

- Designing a zero-trust, dual-mode security architecture for Software-Defined Vehicular Networks, featuring a lightweight $O(1)$ rule-based engine with HMAC authentication for rapid attack filtering.
- Developing an advanced AI pipeline integrating a Temporal Graph Neural Network (TGNN) for spatio-temporal attack detection and a fine-tuned LLM with RAG for contextual threat reasoning.
- Simulating V2X network models in NS-3 and implementing blockchain-based smart contracts for immutable trust verification and secure flow control.

Multi-Agent Abuse Help AI Assistant | Python, LangGraph, FAISS, DistilBERT | [🐙Code](#) · [📺Demo](#)

- Co-developed a 4-agent AI system (Situation Listener, RAG Retriever, Risk Evaluator, Safety Planner) orchestrated with LangGraph to provide evidence-grounded, risk-aware support.
- Implemented RAG using FAISS and MiniLM embeddings alongside a fine-tuned DistilBERT model for multi-class abuse categorization and transformer-based distress detection via few-shot prompting.
- Built a Streamlit web application with offline reproducibility, evaluating performance comprehensively using Accuracy, F1-score, Hit@K, Precision@K, and Faithfulness metrics.

Cloud Native E-Commerce Microservices | Docker, Kubernetes, AWS, RabbitMQ | [🐙Code](#) · [📺Demo](#)

- Architected a 6-service cloud-native backend communicating asynchronously via RabbitMQ event messaging and secured with a JWT-authenticated API Gateway.

- Integrated Redis for high-performance caching, Stripe API for seamless payments, and Google OAuth SSO, utilizing the MERN stack for full-stack application logic.
- Engineered a robust CI/CD pipeline leveraging GitHub Actions and Trivy security scanning to deploy containerized microservices to AWS EC2 via production-ready Kubernetes manifests.

High-Performance Convolution Engine | *C++, CUDA, OpenMP, MPI* | [Code](#)

- Spearheaded a distributed-memory high-performance computing project, implementing hybrid MPI and OpenMP modules for parallel Sobel edge detection on large-scale images.
- Engineered row-wise image partitioning and halo-row exchange utilizing `MPI_Sendrecv`, `MPI_Scatterv`, and `MPI_Gatherv` to optimize inter-node communication.
- Accelerated image processing by developing a CUDA-based solution with shared-memory tiling, benchmarking against serial baselines to achieve a $2\times$ speedup while validating accuracy via RMSE.

Automated Cinnamon Grading System | *Python, OpenCV, Scikit-learn* | [Code](#)

- Engineered a computer vision pipeline to grade Ceylon cinnamon quills, applying Otsu thresholding, Canny edge detection, and morphological operations for robust image segmentation.
- Extracted sophisticated features including geometric properties (skeletonization), HSV color-space metrics, and complex textures using GLCM and Local Binary Patterns (LBP).
- Applied `StandardScaler` normalization and trained SVM and KNN classifiers using stratified cross-validation, generating explainable result visualizations with high precision and recall.

Secure File Transfer Protocol | *RSA-2048, AES-256-GCM, Cryptography* | [Code](#)

- Designed a zero-trust, end-to-end encrypted file sharing protocol engineered to defend against Man-in-the-Middle (MitM) attacks, packet sniffing, and data tampering over untrusted networks.
- Implemented a hybrid cryptographic architecture utilizing RSA-2048 (OAEP) for secure key exchange and AES-256-GCM with secure nonce generation for authenticated data encryption (AAD).
- Secured chunked file transfers against replay attacks using session tokens/salts, enforcing data integrity and non-repudiation via HMAC-SHA256 and RSA-PSS digital signatures.

Education

University of Ruhuna, Faculty of Engineering

BSc (Hons) in Computer Engineering; SGPA: 3.54

Sri Lanka

Mar 2022 – Sep 2026

Sri Sumangala Girls' School

G.C.E. Advanced Level – Physical Science Stream

Panadura, Sri Lanka

2006 – 2019

Competitions & Certifications

Key Competitions: Hackaholics 6.0 (Top 10 Finalist), Innovate with Ballerina (WSO2), Eminence 4.0, IEEEExtreme 18.0, Codesprint 8.

AI & Data Certifications: Supervised Machine Learning (Coursera), AI Builders Lab (Google for Developers), Python Data Analytics (Udemy).

Software Dev Certifications: MERN Stack (LinkedIn), Python (UoM), Flutter & Advanced UI/UX, CSS & SQL (HackerRank).

Leadership & Volunteering

IEEEExtreme 18.0 Proctor Program – Sub-Committee General Team Member

Ad Astra 1.0 (Half Night Camp) – Organizing Committee Member

SEDS Ruhuna – Education Team

References

Dr. Prabath Weerasinghe

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